

Exercise 5 – Technical Design and Implementations

Learning objectives: Apply and learn the impact of architectural decisions

Remember, our project is:

Browser-Based TicTacToe for 2 Players

A detailed description can be found in exercise 1. In addition, use the documentation created in the previous exercises as a basis for the following tasks.

This exercise may require you to do some research on technologies you have not encountered before. That is part of a software architect's job!

Part 1 – Implement the Display of the TicTacToe Playing Field

Implement the initial part of the TicTacToe project according to your architectural design decisions from exercise 3.

When you are done, your TicTacToe should be able to do the following:

1. The webserver can be started and should serve content on port 8080.
2. It should serve the page displaying the (still empty) TicTacToe field.
3. Users should be assigned or be able to select their player.
4. Make sure that unit tests are present. The number of required tests and/or target test coverage depends on your quality objectives
(you may have to make an architectural decision on the amount of testing)

Part 2 – Design Patterns

In your Git repository's `docs` directory: Create a file named `"technical-design.md"`. Write the results of the following tasks into this file. Commit and push it to GitLab once you are done.

1. Given your implementation. Which design patterns (e.g., GRASP or SOLID patterns) did you apply?
Write a short bullet point list with the name of the respective pattern, a link to the source file implementing it, and a **very short** description on how this code implements the pattern.